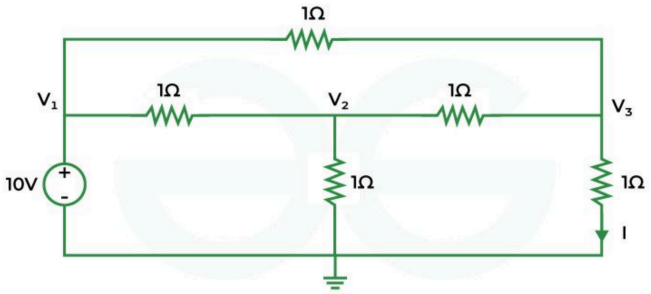
4

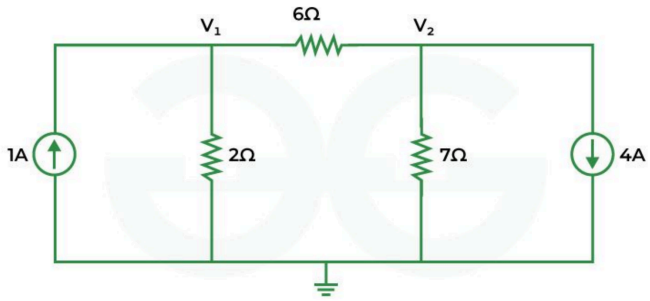
Q. Using nodal analysis find I(With Voltage source)



Ans. 5 A

5

Question .Find V1 and V2 using nodal analysis (With current Source)



Ans. V1 = -2 V and V2 = -14 V

6

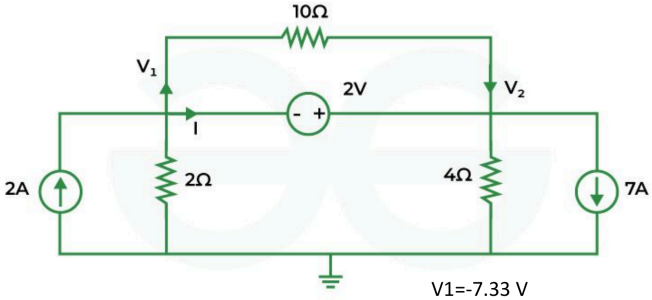
Super Node Analysis

Super node is special condition of nodes. When a voltage source is present between two non-reference node then it is called as super node. KVL and KCL are both applied to the super node to solve the equations.

Properties of Super node

It does not have well defined voltage
Both KCL and KVL are required to solve the equations
The voltage difference is zero between any two nodes in super node

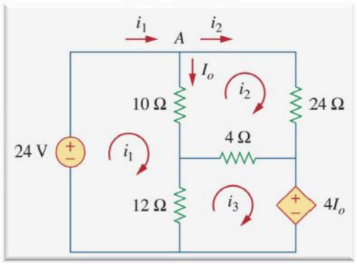
Q. Using nodal analysis find V_1 and V_2



V1=-7.33 V
And V2= -5.33 V

10

Example 3: Use mesh analysis to find the current in the circuit of the shown figure.



We first obtain the mesh currents using KVL. For mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

$$11i_1 - 5i_2 - 6i_3 = 12 \qquad (1)$$

For mesh 2

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

$$-5i_1 + 19i_2 - 2i_3 = 0 \qquad (2)$$

For mesh 3

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

but $I_o = (i_1 - i_2)$

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

$$-i_1 - i_2 + 2i_3 = 0 \qquad (3)$$

Solve equations 1, 2 and 3 to get i_1, i_2 and i_3.

Multiply eq. 2 by 3 and then subtract the resulting eq. from eq.1.

$$11i_1 - 5i_2 - 6i_3 = 12$$

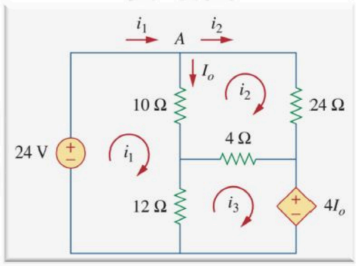
$$-15i_1 + 57i_2 - 6i_3 = 0$$

$$26i_1 - 62i_2 = 12 \qquad (4)$$

8

11

Example 3: Use mesh analysis to find the current in the circuit of the shown figure.



Multiply eq. 3 by 3 and then add the resulting eq. from eq.1.

$$11i_1 - 5i_2 - 6i_3 = 12$$

$$-3i_1 - 3i_2 + 6i_3 = 0$$

$$8i_1 - 8i_2 = 12 \qquad (5)$$

Multiply eq. 5 by 62/8 and then subtract the resulting eq. from eq.4.

$$26i_1 - 62i_2 = 12 \qquad (4)$$

$$62i_1 - 62i_2 = 93$$

$$-36i_1 = -81$$

$$i_1 = 2.25\text{A}$$

Substitute i_1 in eq.4 or eq.5 to get i_2

$$8\times2.25 - 8i_2 = 12$$

$$i_2 = \frac{8\times2.25 - 12}{8} = 0.75\text{A}$$

$$I_o = i_1 - i_2 = 2.25 - 0.75 = 1.5A$$

9

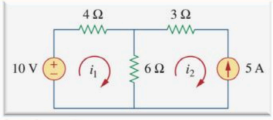
12

Mesh Analysis with Current Sources

CASE 1: When a current source exists only in one mesh.

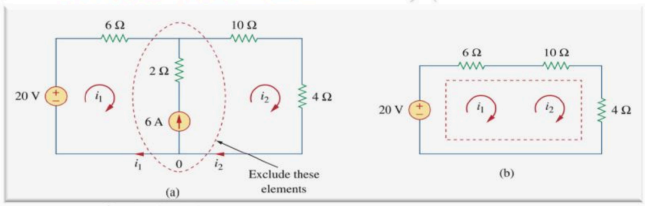
for example. i_2=-5A and write a mesh equation for the other mesh in the usual way; that is,

$$-10 + 4i_1 + 6(i_1 - i_2) = 0 \quad \Rightarrow \quad i_1 = -2\text{ A}$$



CASE 2: When a current source exists between two meshes.

We create a *supermesh* by excluding the current source and any elements connected in series with it.



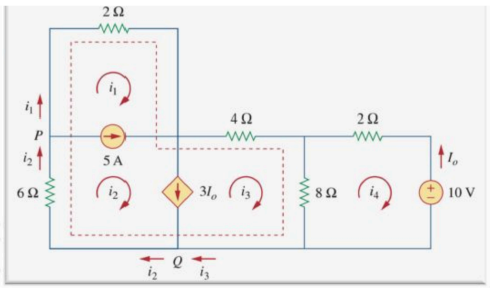
$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

$$i_2 = i_1 + 6$$

A *supermesh* results when two meshes have a (dependent or independent) current source in common.

10

16



For the dependent current source, we apply KCL to node Q:

$i_2 = i_3 + 3I_o$

$I_o = -i_4$

$i_2 = i_3 - 3i_4$ (3)

Applying KVL in mesh 4,

$2i_4 + 8(i_4 - i_3) + 10 = 0$

$5i_4 - 4i_3 = -5$ (4)

Solve these four equation to get mesh currents

$i_1 = -7.5\text{A}$ $i_2 = -2.5\text{A}$

$i_3 = 3.93\text{A}$ $i_4 = 2.143\text{A}$